

Supplementary Material

Full Proofs for the Joint Task Offloading and UAV Trajectory Optimization Problem

This supplementary document provides the full proofs of Theorems 1 and 2 referenced in the main paper: NP-hardness of problem F_1 (Theorem 1) and the generalized ordinal potential property of the task offloading game under bounded externality (Theorem 2). The proof of Theorem 3 is given in the main text. All notation follows the main text.

1 Problem Statement

We consider the joint task offloading and UAV trajectory optimization problem in a Space–Air–Ground Integrated Network (SAGIN) with mobile user equipments (MUEs), UAVs, LEO satellites, and base stations (BSs). Each MUE $e_i \in E$ has a task $Q_i = (A_i, C_i)$ with data size A_i and required CPU cycles C_i . The offloading decision of MUE e_i is a triplet $o_i = (a_i, b_i, c_i)$, where $a_i \in \{0, 1, 2, 3\}$ selects among local computing, UAV, LEO, or BS, b_i is the server index, and c_i is the wireless channel.

Let $\mathcal{V} = U \cup S \cup B$ denote the set of all edge computing nodes. Define $\mathcal{I}_v^+ = \{i \mid a_i^+ \neq 0, v_i^+ = v\}$ as the set of MUEs offloading to node v . The problem is formulated as

$$\min_{(\mathbf{o}^+, \mathbf{q}^+) \in \Omega(t)} \sum_{i=1}^N K_i(o_i^+, \mathbf{o}_{-i}^+, \mathbf{q}^+), \quad (F_1)$$

subject to LEO communication-window constraints, edge-node computation capacity $\sum_{i \in \mathcal{I}_v^+} f_{i,v}^+ \leq F_v^{\max}$ for $v \in \mathcal{V}$, UAV movement $\|q_j^+ - q_j(t)\|_2 \leq D_{\max}$, individual rationality $K_i \leq K_{e_i}^{\text{local}}$, coverage constraints for UAV and BS, and feasibility of the decision variables.

2 Full Proof of Theorem 1 (NP-hardness)

Theorem 1. *The optimization problem F_1 is NP-hard.*

Proof. We prove NP-hardness by showing that the Generalized Assignment Problem (GAP), a classical NP-hard problem, is a restricted instance of F_1 . Consider a special case of F_1 where UAV positions are fixed, local computing is disabled, channel selection and co-channel interference are ignored, and each MUE must offload to exactly one feasible edge server. Candidate servers include UAVs, LEO satellites, and BSs, each with finite computation capacity.

Given any GAP instance, each job maps to one MUE task and each machine maps to one edge server in $\mathcal{V} = U \cup S \cup B$. Let $x_{ij} \in \{0, 1\}$ denote whether task i is assigned to server j . Assigning task i to server j incurs offloading cost K_{ij} (GAP assignment cost) and consumes r_{ij} units of computation resource. Since UAV positions are fixed and wireless interference is ignored, K_{ij} is

constant. Each server j has capacity R_j . Assignments violating LEO visibility, UAV coverage, or BS coverage constraints are infeasible.

Under this restricted setting, the problem becomes

$$\begin{aligned}
& \min_{\{x_{ij}\}} \quad \sum_{i=1}^N \sum_{j \in \mathcal{V}} x_{ij} K_{ij} \\
& \text{s.t.} \quad \sum_{j \in \mathcal{V}} x_{ij} = 1, \quad \forall i, \\
& \quad \sum_{i=1}^N x_{ij} r_{ij} \leq R_j, \quad \forall j \in \mathcal{V}, \\
& \quad x_{ij} = 0, \quad \text{if assignment } (i, j) \text{ is infeasible,} \\
& \quad x_{ij} \in \{0, 1\}, \quad \forall i, j.
\end{aligned} \tag{1}$$

This is a generalized assignment problem with forbidden assignments. Since GAP is NP-hard, this restricted instance of F_1 is NP-hard. The original F_1 further includes local-computing decisions, channel selection, co-channel interference, LEO visibility, UAV/BS coverage, and UAV trajectory optimization, so F_1 is at least as general and NP-hard. $\square$

3 Full Proof of Theorem 2 (Generalized Ordinal Potential Game)

For a fixed UAV deployment $\mathbf{q}$, task offloading is modeled as a finite non-cooperative game $\mathcal{G} = \{\mathcal{N}, (\mathcal{O}_i)_{i \in \mathcal{N}}, (K_i)_{i \in \mathcal{N}}\}$, where $\mathcal{N} = \{1, \dots, N\}$ is the set of MUEs, $\mathcal{O}_i$ is the finite feasible strategy set of MUE i , and $K_i(o_i, \mathbf{o}_{-i}; \mathbf{q})$ is its delay-energy cost. The strategy profile is $\mathbf{o} = (o_1, \dots, o_N)$. The bounded-externality condition requires that each accepted profitable update in the best-response dynamics satisfies

$$\Delta K_{-i} \leq \rho |\Delta K_i|, \quad 0 \leq \rho < 1, \tag{2}$$

where ΔK_i is the updating MUE's cost variation and ΔK_{-i} is the induced variation of the other MUEs.

Theorem 2. *For any fixed UAV deployment $\mathbf{q}$, suppose each accepted profitable update in the BRG dynamics satisfies the bounded-externality condition. Then $\mathcal{G}$ is a generalized ordinal potential game admitting at least one pure-strategy Nash equilibrium, and the sequential best-response dynamics in BRG converges to a pure-strategy Nash equilibrium in finitely many iterations.*

Proof. Define the potential function as total system cost: $\Phi(\mathbf{o}; \mathbf{q}) = \Psi(\mathbf{o}; \mathbf{q}) = \sum_{j=1}^N K_j(o_j, \mathbf{o}_{-j}; \mathbf{q})$.

Consider a unilateral update of MUE i from o_i to o'_i within $\mathcal{O}_i$, while others remain fixed. Let

$$\Delta K_i = K_i(o'_i, \mathbf{o}_{-i}; \mathbf{q}) - K_i(o_i, \mathbf{o}_{-i}; \mathbf{q}), \tag{3}$$

and

$$\Delta K_{-i} = \sum_{j \neq i} [K_j(o_j, o'_i, \mathbf{o}_{-\{i,j\}}; \mathbf{q}) - K_j(o_j, o_i, \mathbf{o}_{-\{i,j\}}; \mathbf{q})]. \tag{4}$$

Then the change in the potential function is

$$\begin{aligned}
\Delta \Phi &= \Phi(o'_i, \mathbf{o}_{-i}; \mathbf{q}) - \Phi(o_i, \mathbf{o}_{-i}; \mathbf{q}) \\
&= [K_i(o'_i, \mathbf{o}_{-i}; \mathbf{q}) - K_i(o_i, \mathbf{o}_{-i}; \mathbf{q})] \\
&\quad + \sum_{j \neq i} [K_j(o_j, o'_i, \mathbf{o}_{-\{i,j\}}; \mathbf{q}) - K_j(o_j, o_i, \mathbf{o}_{-\{i,j\}}; \mathbf{q})] \\
&= \Delta K_i + \Delta K_{-i}.
\end{aligned} \tag{5}$$

In BRG, an update is accepted only if profitable, i.e., $\Delta K_i < 0$. Under the bounded-externality condition $\Delta K_{-i} \leq \rho |\Delta K_i|$, $0 \leq \rho < 1$. Since $\Delta K_i < 0$, $|\Delta K_i| = -\Delta K_i$. Thus,

$$\Delta \Phi = \Delta K_i + \Delta K_{-i} \leq \Delta K_i + \rho |\Delta K_i| = -(1 - \rho) |\Delta K_i| < 0. \quad (6)$$

Thus every profitable unilateral update strictly decreases Φ , so Φ is a generalized ordinal potential function.

Since each MUE has a finite strategy set, $\prod_{i \in \mathcal{N}} \mathcal{O}_i$ is finite, and Φ takes finitely many values. Because every accepted update strictly decreases Φ , the best-response process cannot cycle and terminates in finitely many iterations.

Thus $\mathbf{o}^*$ satisfies

$$K_i(o_i^*, \mathbf{o}_{-i}^*; \mathbf{q}) \leq K_i(o_i, \mathbf{o}_{-i}^*; \mathbf{q}), \quad \forall i \in \mathcal{N}, \forall o_i \in \mathcal{O}_i. \quad (7)$$

This is a pure-strategy Nash equilibrium. Therefore, under bounded externalities, the task offloading game admits at least one pure-strategy Nash equilibrium, and the sequential BRG algorithm converges in finitely many iterations. $\square$